240118: metamonzogranite, Mount Mason

(Yilgarn Craton granites, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Menzies (SH 51-5), Mount Mason (2939)
MGA Zone 51, 245341E 6772951N

Sampled on 5 May 2019

This sample was collected from a low outcrop, about 12.9 km west of Spotted Dog Hill, 9.4 km north-northwest of Mount Ida, and 3.5 km southeast of Mount Mason.

Geological context

The unit sampled is a metamonzogranite adjacent to the western margin of the Mount Ida greenstone belt in the eastern Youanmi Terrane (Wyche and Duggan, 2003). The metamonzogranite is interpreted to postdate deformation in the adjacent greenstones (Zibra et al., 2020). A weathered, foliated monzogranite sampled 10.3 km to the west-southwest provided an igneous crystallization age of 2693 ± 5 Ma (sample GA 96969016, Bodorkos, 2011). A porphyritic metamicrogranite collected about 14.3 km to the southeast yielded an igneous crystallization age of 2700 ± 9 Ma (GSWA 169078, Nelson, 2002).

Petrographic description

The sample is a medium- to coarse-grained biotite metamonzogranite (Fig. 1), consisting of about 35–40% K-feldspar, 30% plagioclase, 25% quartz, 5% myrmekite, 3% biotite, <1% magnetite–hematite and accessory muscovite, chlorite, apatite, allanite, fluorite and zircon. K-feldspar is anhedral, equant to tabular, microperthitic, and up to 10 mm in size. Some K-feldspar grains show microcline twinning, some contain small plagioclase inclusions, and some are recrystallized along fractures and grain margins. Plagioclase is subhedral, mostly equant, up to 5 mm in diameter, is partially altered to sericite or clay minerals, and some crystals are compositionally zoned. Quartz forms aggregates or lenses up to 12 mm long of microcrystalline recrystallized grains, which have strongly serrated, intergrown boundaries, and exhibit undulose extinction. Myrmekite forms invasive lobes and aggregates up to 1.5 mm in diameter at the margins of larger K-feldspar grains. Myrmekite also forms irregular aggregates up to 2.5 mm in size that consist of many small myrmekite lobes that have replaced entire K-feldspar grains. Biotite forms anhedral to subhedral, equant grains and laths up to 1 mm in size, many in small aggregates, and is partially altered to chlorite. Muscovite occurs with biotite or in the cores of some plagioclase grains. Magnetite is anhedral to subhedral, up to 0.7 mm in size, and partly altered to hematite. Apatite forms euhedral laths up to 0.15 mm long, and pale brown metamict allanite is up 0.25 mm diameter. Fine-grained purple fluorite is interstitial to fine-grained magnetite. The sample lacks a tectonic foliation, although strained grains, microcline twinning, myrmekite development, and recrystallization suggest deformation and recovery at moderate to high-grade metamorphic conditions.

Zircon morphology

Zircons isolated from this sample are pale brown to dark brown or opaque and mainly subhedral to euhedral. The crystals are up to 350 µm long, and mostly elongate, with aspect ratios up to 9:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and most crystals are dominated by high-U, metamict zones. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 13–14 August 2020, using SHRIMP-B. Ten analyses of the M257 standard were obtained during the session, of which eight analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.61% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.32% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Seventeen analyses were obtained from 17 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to strongly discordant (Fig. 3). Three analyses are >20% discordant. The dates obtained from these three analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 14 analyses form a single group, based on their $^{207}Pb^{*}/^{206}Pb^{*}$ ratios.

Figure 1. Hand specimen image of sample 240118: metamonzogranite, Mount Mason

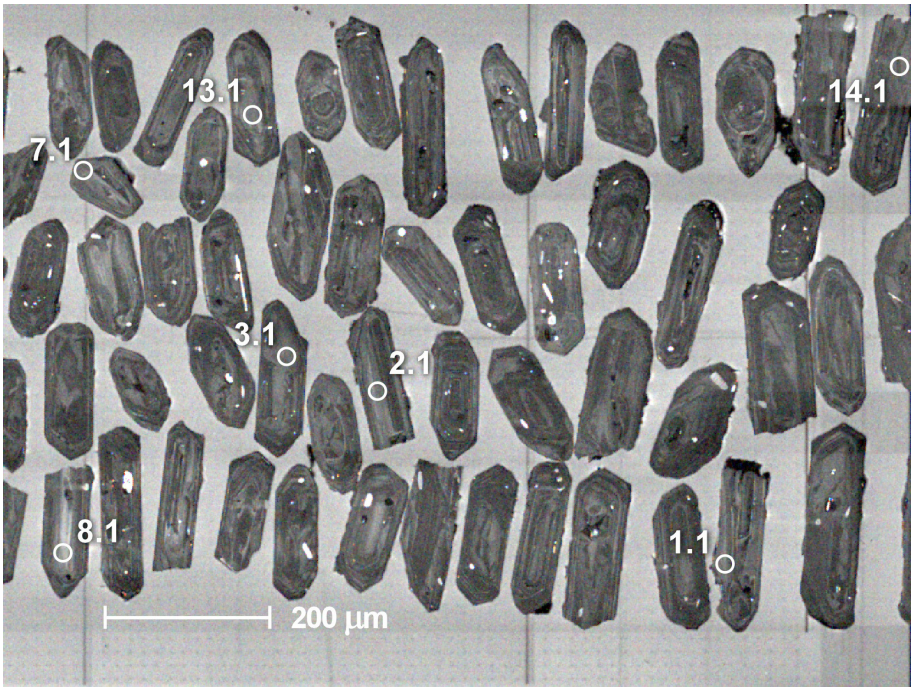

Figure 2. Cathodoluminescence image of representative zircons from sample 240118: metamonzogranite, Mount Mason. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 240118: metamonzogranite, Mount Mason

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	14	14.1	230	41	0.18	0.258	1.960	0.021	0.18469	0.00093	1.965	0.021	0.18238	0.00106	2652	24	2675	10	0.9
I	3	3.1	198	51	0.26	0.146	2.034	0.022	0.18408	0.00104	2.037	0.022	0.18278	0.00112	2575	24	2678	10	3.8
I	16	16.1	100	84	0.84	0.145	1.990	0.028	0.18425	0.00150	1.993	0.028	0.18295	0.00161	2621	31	2680	15	2.2
I	9	9.1	253	390	1.55	0.196	1.912	0.020	0.18471	0.00089	1.916	0.020	0.18297	0.00099	2708	24	2680	9	-1.0
I	11	11.1	165	229	1.39	0.388	2.225	0.036	0.18673	0.00118	2.234	0.036	0.18327	0.00140	2385	33	2683	13	11.1
I	12	12.1	96	81	0.84	-0.055	1.932	0.027	0.18291	0.00141	1.931	0.027	0.18340	0.00145	2690	31	2684	13	-0.2
I	15	15.1	170	47	0.27	0.324	1.979	0.023	0.18630	0.00106	1.985	0.023	0.18341	0.00124	2630	25	2684	11	2.0
I	4	4.1	276	50	0.18	0.162	1.890	0.019	0.18533	0.00089	1.893	0.019	0.18389	0.00097	2735	23	2688	9	-1.7
I	5	5.1	252	55	0.22	0.033	1.938	0.020	0.18425	0.00090	1.938	0.020	0.18395	0.00092	2682	23	2689	8	0.3
I	8	8.1	79	49	0.62	0.000	1.950	0.029	0.18415	0.00165	1.950	0.029	0.18415	0.00165	2669	33	2691	15	0.8
I	2	2.1	172	102	0.59	0.566	2.057	0.025	0.18990	0.00122	2.068	0.025	0.18485	0.00155	2542	26	2697	14	5.7
I	13	13.1	162	66	0.41	-0.017	2.111	0.025	0.18472	0.00113	2.111	0.025	0.18487	0.00114	2500	25	2697	10	7.3
I	10	10.1	894	391	0.44	0.023	1.926	0.016	0.18591	0.00078	1.926	0.016	0.18570	0.00078	2696	19	2705	7	0.3
I	6	6.1	308	73	0.24	0.376	2.001	0.020	0.18978	0.00091	2.008	0.020	0.18643	0.00116	2605	22	2711	10	3.9
D	17	17.1	746	408	0.55	0.179	2.806	0.039	0.16314	0.00155	2.811	0.040	0.16154	0.00158	1962	24	2472	16	20.6
D	7	7.1	196	151	0.77	1.231	3.622	0.084	0.19452	0.00138	3.668	0.085	0.18354	0.00223	1554	32	2685	20	42.1
D	1	1.1	187	111	0.59	0.904	2.758	0.062	0.19175	0.00104	2.783	0.063	0.18368	0.00158	1979	39	2686	14	26.3

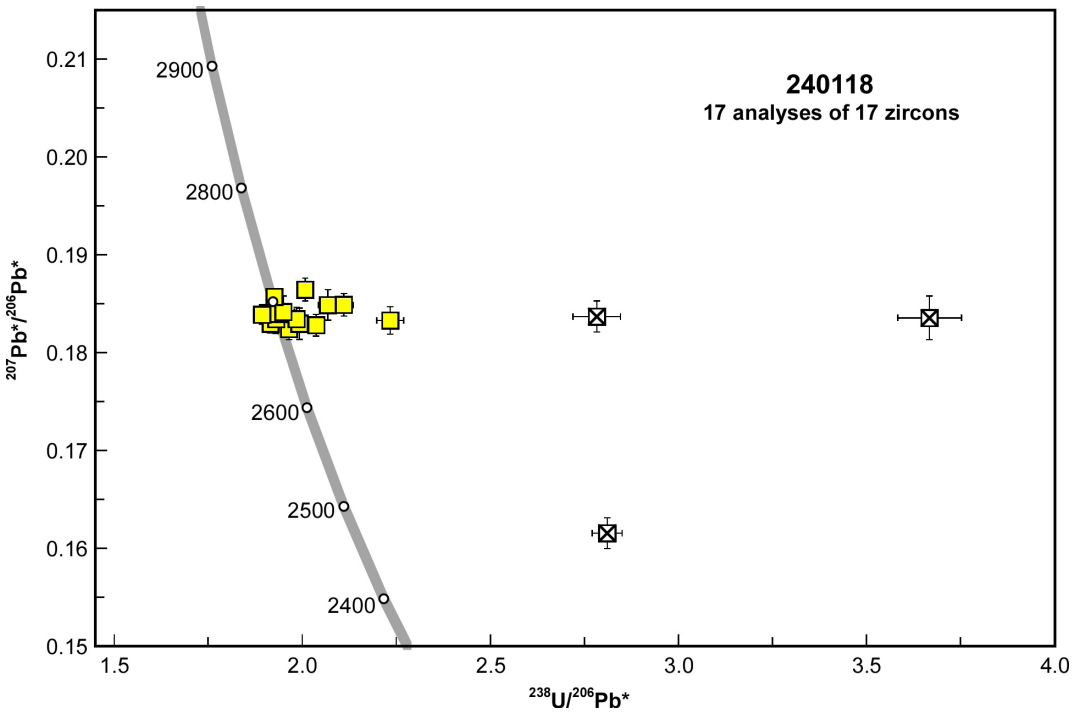

Figure 3. U–Pb analytical data for sample 240118: metamonzogranite, Mount Mason. Yellow squares indicate Group I (magmatic zircons); crossed squares indicate Group D (discordance >20%)

Group I comprises 14 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2690 ± 7 Ma (MSWD = 1.2).

The date of 2690 ± 7 Ma for the 14 analyses in Group I is interpreted as the igneous crystallization age of the monzogranite.

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Zibra, I 2021, 240118: metamonzogranite, Mount Mason; Geochronology Record 1832: Geological Survey of Western Australia, 4p.

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Data released: 10 December 2021